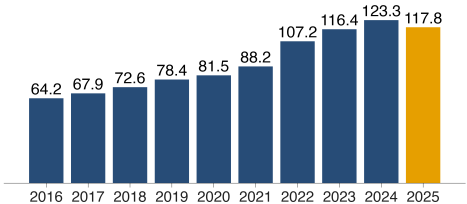
Market Integration and the Effects of Displacement on Host Communities

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Forced Displacement



The number of displaced people in the world has almost doubled in the last decade

Displacement is a shock on two markets for host communities

- A demand shock on consumption markets
- A supply shock on labor markets

} **Effects on host communities will depend on the nature and intensity of each shock**

1) How does displacement affect labor and consumption markets in host communities?

Market integration and displacement

Rural areas host a substantial share of displaced people

- Difficult to estimate: 60% of internally displaced people (IDPs) and 30% of refugees (UNHCR, 2024)

Rural areas of LMICs are often poorly integrated with external markets

- Sub-Saharan Africa: distance to paved road >10 km outside of major cities (Djemaï et al., 2024)
- Market integration provides economic opportunities outside of agriculture (e.g., Asher and Novosad (2020); Gertler et al. (2024))

Market integration might affect how host communities respond to displacement inflows

- **Integrated areas**: dilution of the demand shock and sectoral reallocation out of IDP-exposed jobs is plausible
- **Less integrated areas**: not clear with isolated markets + absence of off-farm alternatives

2) Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

Overview of the project

1) How does displacement affect labor and consumption markets in host communities?

- Study Ethiopia's 2018–2022 displacement crisis
- Build a stylized framework about the effect of displacement on labor and consumption markets
- Merge Ethiopian Socioeconomic Survey (ESS) panel data with IOM Displacement Tracking Matrix (DTM) data and ACLED conflict data
- Effect on outcomes of host communities based on short-term exposure to displacement
- Use a shift-share IV design with an "enclave" instrument

2) Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

- Adapt stylized framework to market integration
- Construct an index of market integration: distance to trade and market amenities
- Estimate heterogeneous effects of displacement in areas with high and low market integration

Preview of results

1) How does displacement affect labor and consumption markets in host communities?

- **Labor markets**:
  - Wages fall in low-skill occupations
  - Hosts experience occupational upgrading in the form of sectoral reallocation
- **Consumption markets**:
  - Local service prices fall
  - Slight decrease in food prices

Consistent with displacement as a labor supply shock dominating the demand shock

Preview of results

2) Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

High market integration

- Hosts move toward self- and wage employment
- Small reduction in prices

Low market integration

- No sectoral reallocation
- Agricultural production increase
- All prices decrease

Sustained/increased well-being with different patterns of adjustment:

Sectoral reallocation

Expansion within agriculture

Contributions to the literature

1) **Displacement and migration**: null/positive effects for the hosts and occupational upgrading (Card, 1990; Cortes, 2008; Alix-Garcia and Saah, 2010; Alix-Garcia and Bartlett, 2015; Calderón-Mejía and Ibáñez, 2016; Balkan and Tumen, 2016; Foged and Peri, 2016; Borjas and Monras, 2017; Morales, 2018; Alix-Garcia et al., 2018; Altındağ et al., 2020; Rozo and Winkler, 2021; Derenoncourt, 2022; Assaad et al., 2023; Winton, 2025; Bohsali, 2025; Imbert and Ulyssea, 2026)
   - Adjustment varies with market integration
   - In less integrated areas, displacement inflows leads to expansion within agriculture
   - Displacement can lead to overall price reductions, even in remote settings

2) **Market integration:** transportation infrastructure expands economic opportunities (Dercon et al., 2009; Gollin and Rogerson, 2014; Atkin and Donaldson, 2015; Donaldson and Hornbeck, 2016; Donaldson, 2018; Asher and Novosad, 2020; Brooks and Donovan, 2020; Shamdasani, 2021; Gertler et al., 2024; Djemaï et al., 2024; Adamopoulos, 2025)
   - Market integration shapes adjustment of markets to displacement
     - Labor: opportunities for the hosts in off-farm employment
     - Consumption: absorb price effects of additional demand from the displaced

3) **Structural transformation:** productivity gaps and barriers to reallocating labor out of agriculture (Lagakos and Waugh, 2013; Gollin et al., 2014; Gollin and Rogerson, 2014; Bryan et al., 2014; Bryan and Morten, 2019; Lagakos, 2020; Lagakos et al., 2020, 2023; Donovan et al., 2023; Gollin and Kaboski, 2023; Baseler, 2023)
   - Sectoral reallocation caused by displacement inflows requires integrated markets in this context
   - Gains in living standards with expansion within agriculture

1) How does displacement affect labor and consumption markets in host communities?

Stylized framework with full market integration

Production: food F and local services S

$$Y_F = A_F \, \bar{T}_F^{1-\alpha} \, {L_F}^{\alpha}, \qquad Y_S = A_S \, {L_S}^{\beta} \, {H_S}^{1-\beta}$$

- L: low-skill labor, H: high-skill labor
- A: sectoral technology, $\bar{T}_F$: fixed agricultural land
- Food is tradable at world price $\bar{p}_F$ and services are nontradable

Consumption: Cobb–Douglas over food and services, fixed spending

$$U_j = c_F^{\omega_F} \, c_S^{\omega_S}, \qquad p_F c_{Fj} + p_S c_{Sj} \leq y_j$$

- Partial equilibrium: financing of food imports is outside the model

Roy-style selection: hosts choose occupations by comparative advantage $\theta_i \overset{\text{iid}}{\sim} G(\theta_i)$

$$\text{Choose high-skill work if} \quad \theta_i \, w_H \geq w_L \quad \Longleftrightarrow \quad \theta_i \geq \Theta_0 = \frac{w_L}{w_H}.$$

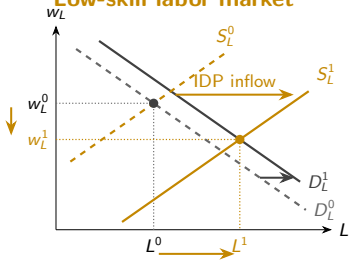
$\Rightarrow$ **Displacement adds low-skill workers and consumers**

Labor markets

IDP workers added $>$ service jobs demanded

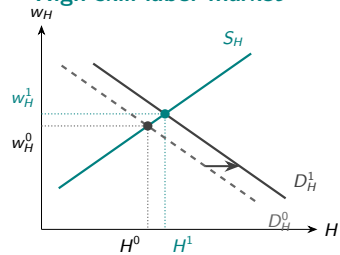
$w_L \searrow$ $w_L \nearrow$

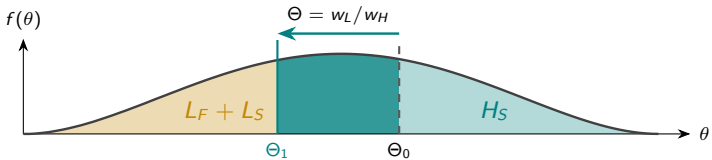
Low-skill labor market



1) $w_L/w_H \searrow$

High-skill labor market





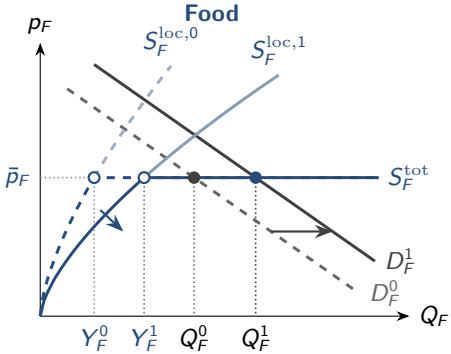
As w_L/w_H decreases
hosts leave L_F and L_S for H_S

2) Hosts move towards high-skill occupations

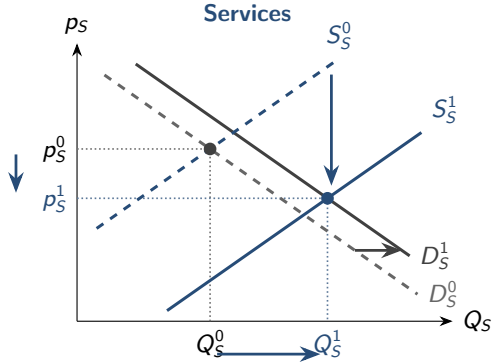
Consumption markets

IDP workers added $>$ service jobs demanded

$w_L \searrow$ $w_L \nearrow$



4) Food prices unaffected



3) Decrease in price of local services

Model predictions

If low-skill labor supply effects dominate demand effects:

1. **Low-skill to high-skill wage ratio falls**
2. **Hosts move toward high-skill occupations**
3. **Prices for local services fall**
4. **Food prices are unchanged under full integration**

Setting: Ethiopia

Labor markets and annual consumption

- 51% household agriculture, 42% local non-tradable, 7% other
- Annual adult-equivalent consumption per capita: \$516 (2021 PPP) $\Rightarrow$ \$1.4/day
- 24% annual inflation 2018-2022

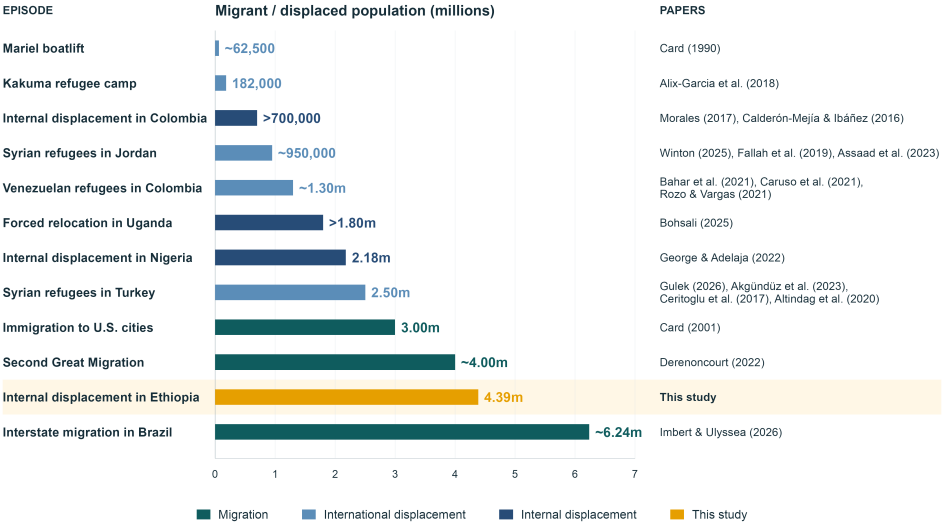
Conflict in Ethiopia

- After two decades of stability and rapid economic growth, inter-communal violence broke off in 2016
- By 2018, highest number of new IDPs ever registered in a single country over a year: 2.9 mio (IDMC, 2019)
- Cumulative new displacements reached 5.1 mio by 2021 (IDMC, 2022)

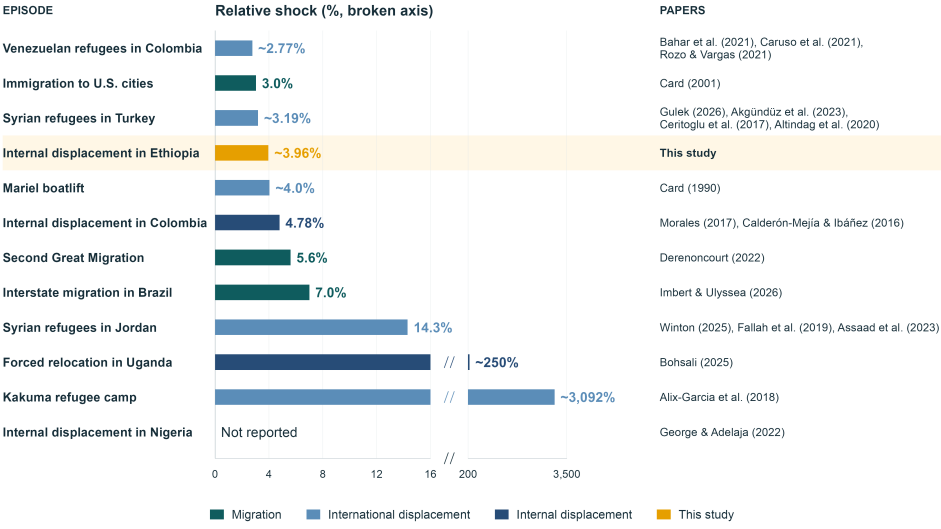
2 types of conflicts

- Inter-ethnic conflict (2016 - present): whole country
- Tigray war (2020-present): North

The Ethiopian displacement crisis is one of the largest ever studied



But is similar to the Mariel boatlift in relative terms



What do IDPs do?

Difficult to have precise information on IDPs

- DTM data has information on 'majority of IDPs in this site' only
- 20% of sites report having access to income-generating activities
- Main occupations among those reporting access: agriculture (20%), daily labor (45%), petty trade (6%)
- Main source of food: food aid (70%), host donation (9%), self-reliance (6%)

IOM 2025 reports using HH-level data from 3 regions (IOM, 2025)

- Casual labor is is the main source of income for 43% of IDP households

⇒ **Most IDP sites are dominated by households working in casual labor and agriculture**
⇒ **Important reliance on food aid**

Overview of the estimation strategy

Main outcomes: labor market and consumption outcomes for the hosts

- Hours of labor in the last 7 days by sector
- Food and non-food prices in markets of host communities

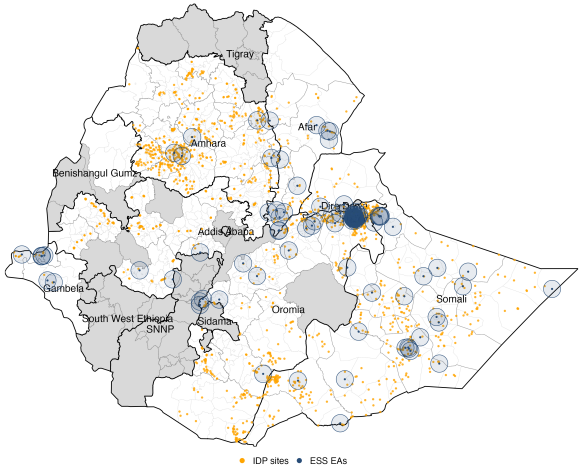
Treatment: short-term exposure to displacement flows

- Annual change in the number of IDPs living near host communities

Instrument: shift-share that predicts exposure to displacement

- **Shares:** before the shock, what share of IDPs in each destination came from each origin district?
- **Shifts:** subsequent conflict events in those origin districts

Treatment variable: exposure to IDPs

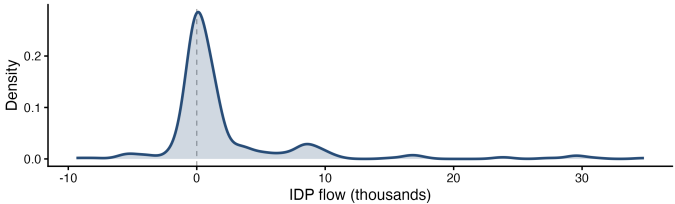


▸ Sum. Stats HH sample

Treatment is the net IDP flow in the last year within 25 km of host communities

$$D_{at} = \sum_{d \in \mathcal{D}_{at}^{25}} \left(IDP_{dt} - IDP_{dt_{-1}} \right)$$

Treatment variable summary statistics - household sample



Variable	Observations	Mean	SD	Min	Max
IDP flow within 25 km (1,000s)	4046	2.44	6.17	-9.37	35.12
Baseline IDPs within 25 km (1,000s)	4046	10.53	10.81	0.07	103.60
Population within 25 km (1,000s)	4046	1299.23	1529.18	6.43	4447.41
IDP flow / population within 25 km (%)	4046	0.53	3.30	-13.84	38.74
Total IDPs / population within 25 km (%)	4046	5.65	10.89	0.01	81.66

Treatment variable

Estimating equation

The effect of internal displacement on host households is estimated according to:

$$y_{ihat} = \beta\ D_{at} + \mathbf{X}'_{ihat}\Pi + \lambda_a + \delta_t + \varepsilon_{ihat} \tag{1}$$

where:

- y_{ihat} is an outcome variable for individual i in household h and enumeration area a at time t
- D_{at} is exposure to net displacement flows
- $\mathbf{X}'_{iht}$ is a vector of control variables[1]
- λ_a and δ_t are enumeration area (EA) and time fixed effects

But OLS would confound the effect of displacement (e.g. positive selection bias)

[1] household size, age, gender, literacy, school attendance, primary school completion, number of conflict events within 25 km, number of conflict events with fatalities within 25 km, number of events with civilian targeting within 25 km

Instrument

Shift-share instrumental variable design:

- "Enclave" logic: use tendency of IDPs to move towards members of their former community
- Evidence that IDPs in Ethiopia move along ethnic lines

$$Z_{at} = \sum_{o} s_{ao} \times \widetilde{c}_{ot} \tag{2}$$

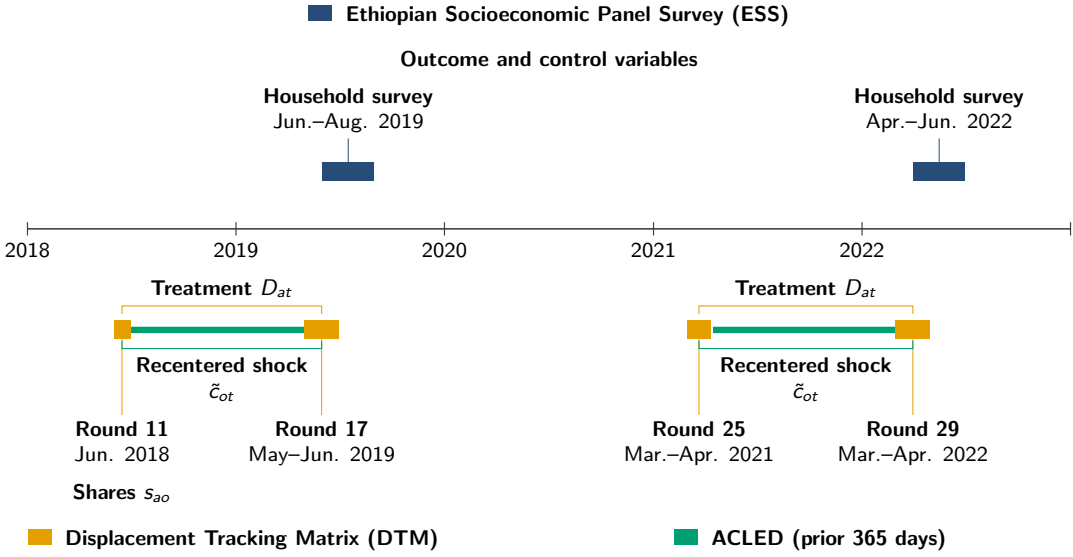
$$\widetilde{c}_{ot} = c_{ot} - \bar{c}_{t} \tag{3}$$

- s_{ao}: pre-shock share of IDPs around enumeration area a that originate from district o
- $\tilde{c}_{ot}$: number of conflict events recorded in district o over the year preceding the observation at t, as a deviation from the mean

Interpretation:

- A destination receives more IDPs when the districts of origin of its previously displaced population experience more conflict

Timeline


Ethiopian Socioeconomic Panel Survey (ESS)
Outcome and control variables
Household survey
Jun.–Aug. 2019
Household survey
Apr.–Jun. 2022
2018
2019
2020
2021
2022
Treatment D_{at}
Recentered shock
$\tilde{c}_{ot}$
Round 11
Jun. 2018
Round 17
May–Jun. 2019
Treatment D_{at}
Recentered shock
$\tilde{c}_{ot}$
Round 25
Mar.–Apr. 2021
Round 29
Mar.–Apr. 2022
Shares s_{ao}
Displacement Tracking Matrix (DTM)
ACLED (prior 365 days)

Identification and clustering

Identifying assumptions

- Shift-level exogeneity: origin conflict must be as good as random for host communities
- Affects host outcomes through displacement
- Fixed effects absorb persistent differences and common time shocks

Threats: other changing links between origins and destinations

- **Conflict spillovers:** control for destination conflict; only 19 households report a conflict shock in the past year.
- **Trade disruption:** a trade disruption channel predicts higher prices and lower deflated consumption, effects go the other way.

Shared dependence to common conflict shocks

- Host households share shock through common baseline enclaves
- SEs clustered at the district of origin through an origin-level dataset (Borusyak et al., 2022)

First stage statistics

	IDP flow within 25km (thousands)	
	Household FE	Enumeration-area FE
Recentered conflict-event shock	0.24***	0.14***
	(0.06)	(0.04)
BH 2023 RI p-value	0.038	0.046
Kleibergen–Paap rk Wald F	13.54	13.33
KP rk Wald F p-value	4.16×10^{-4}	4.25×10^{-4}
KP rk Wald F degrees of freedom	1, 82	1, 96
Household-year observations	2,896	4,145
Households	1,448	2,697
Origin-year cells	166	180
Effective shocks	14.71	7.09
Origin-woreda clusters	83	97

How does displacement affect labor and consumption markets in host communities?

If low-skill labor supply effects dominate demand effects:

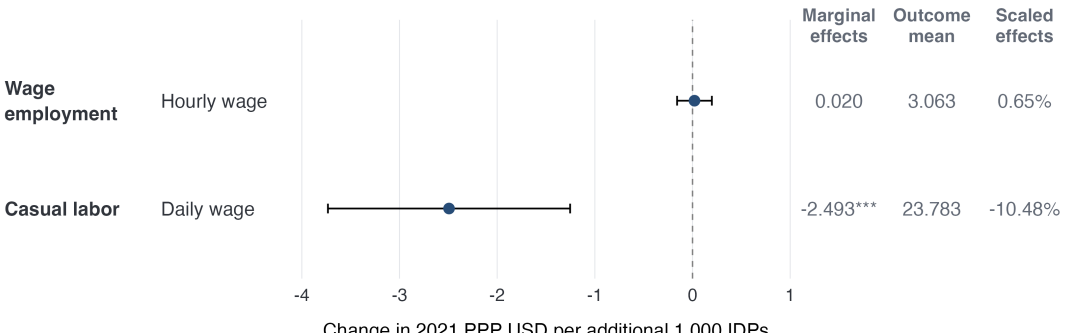
1. **Low-skill to high-skill wage ratio falls**
2. **Hosts move toward high-skill occupations**
3. **Prices for local services fall**
4. **Food prices are unchanged under full integration**

How does displacement affect labor and consumption markets in host communities?

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1) Low-skill to high-skill wage ratio falls



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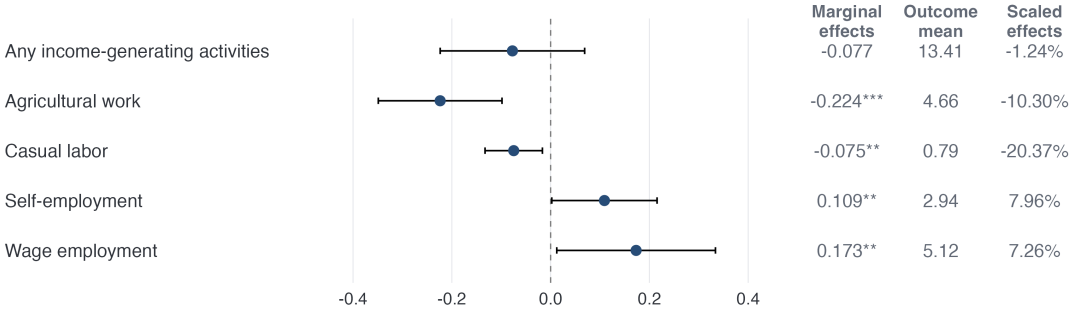
1. **Low-skill to high-skill wage ratio falls**
   Casual labor wage ↘; no effect on permanent employment wage

How does displacement affect labor and consumption markets in host communities?

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1. **Low-skill to high-skill wage ratio falls**
   Casual labor wage ↘; no effect on permanent employment wage
2. **Hosts move toward high-skill occupations**

2) Hosts move toward high-skill occupations



	Marginal effects	Outcome mean	Scaled effects
Any income-generating activities	-0.077	13.41	-1.24%
Agricultural work	-0.224***	4.66	-10.30%
Casual labor	-0.075**	0.79	-20.37%
Self-employment	0.109**	2.94	7.96%
Wage employment	0.173**	5.12	7.26%

How does displacement affect labor and consumption markets in host communities?

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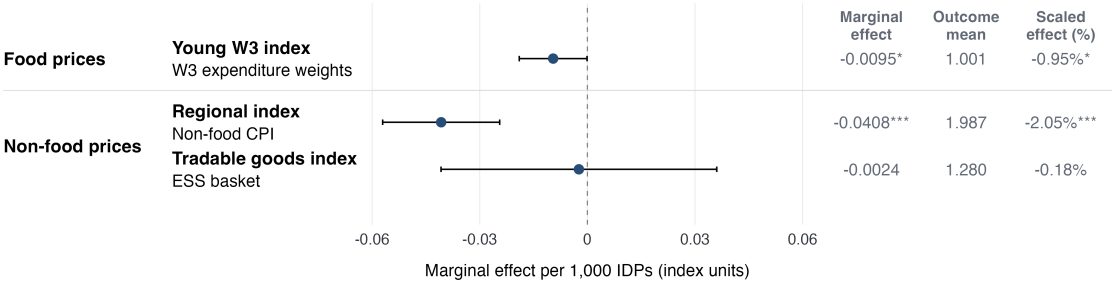
1. **Low-skill to high-skill wage ratio falls**
   Casual labor wage $\searrow$; no effect on permanent employment wage
2. **Hosts move toward high-skill occupations**
   Hosts leave agricultural and casual labor $\Rightarrow$ self- and wage employment

How does displacement affect labor and consumption markets in host communities?

If low-skill labor supply effects dominate demand effects:

1. **Low-skill to high-skill wage ratio falls**
   Casual labor wage $\searrow$; no effect on permanent employment wage
2. **Hosts move toward high-skill occupations**
   Hosts leave agricultural and casual labor $\Rightarrow$ self- and wage employment
3. **Prices for local services fall**
4. **Food prices are unchanged under full integration**

3) Prices for local services fall; 4) Food prices are unchanged



How does displacement affect labor and consumption markets in host communities?

If low-skill labor supply effects dominate demand effects:

1. **Low-skill to high-skill wage ratio falls**
   Casual labor wage $\searrow$; no effect on permanent employment wage
2. **Hosts move toward high-skill occupations**
   Hosts leave agricultural and casual labor $\Rightarrow$ self- and wage employment
3. **Prices for local services fall**
   Prices for non-food falls 2%; but null effect on tradable prices
4. **Food prices are unchanged under full integration**
   Food prices decrease slightly ($< 1\%$)

2) Does the effect of displacement on labor and consumption markets in host communities differ by market integration?

Adding market integration into the framework

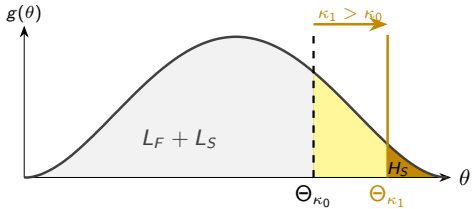
1) Labor markets

Few off-farm opportunities with low integration

- Commuting cost κ for H_S:

$$r_H(\theta_i) = \theta_i w_H - \kappa \bar{c}$$

- Roy cutoff: $\Theta_\kappa = \frac{w_L + \kappa \bar{c}}{w_H}$
- Share of workers in H_S $\searrow$ with κ



Limited reallocation due to lower density at Θ_{κ_1}

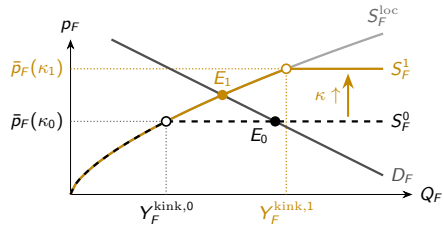
2) Food sector

Isolated markets clear internally

- Iceberg cost κ on imported food:

$$\bar{p}_F = (1 + \kappa) p^*$$

- Market integration governs at what level world prices p^* kick in



Food prices adjust locally

Model predictions

If low-skill labor supply effects dominate demand effects:

High market integration

1. Low-skill to high-skill wage ratio falls
2. Hosts move towards high-skill occupation
3. Prices for local services fall
4. Food prices are unchanged

Low market integration

1. –
2. Limited reallocation
3. –
4. Prices adjust locally

Defining market integration

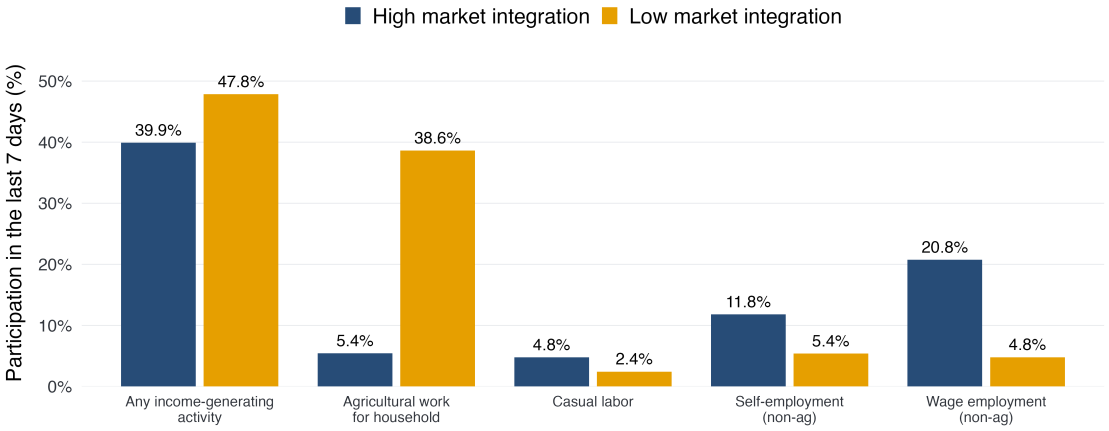
Survey-based measures

- Distance to nearest asphalt road
- Distance to nearest bus station
- Distance to nearest large weekly market

Variable	N	Mean	Median	SD	Min	Max
Nearest asphalt road (km)	4,046	17.92	1.50	41.21	0.00	397.50
Nearest bus station (km)	3,989	8.94	3.00	31.36	0.00	339.00
Nearest large weekly market (km)	4,046	5.08	0.00	11.34	0.00	92.50
Market integration index	**4,046**	**10.55**	**3.00**	**20.17**	**0.00**	**185.00**
Harris market access	3,989	159.86	59.64	170.48	8.93	1,105.49
Urban status (=1)	4,046	0.60	1.00	0.49	0.00	1.00

Sample split by below/above median ⇒ High/low market integration

Host economies look very different by market integration



⇒ **Household agriculture is by far the main activity in less integrated areas**

Empirical strategy for heterogeneous effects

- Previous estimating equation:

$$y_{ihat} = \beta\ D_{at} + \mathbf{X}'_{ihat}\Pi + \lambda_a + \delta_t + \varepsilon_{ihat}$$

- Transformed to have slope coefficients for the effects of displacement in **high** and **low** market integration areas (with $\mathrm{L}_a \equiv 1 - \mathrm{H}_a$)

$$y_{ihat} = \beta_H\ (D_{at} \times \mathrm{H}_a)\ +\ \beta_L\ (D_{at} \times \mathrm{L}_a)\ +\ \mathbf{W}'_{ihat}\Gamma\ + \lambda_a + \delta_t + \varepsilon_{ihat} \tag{4}$$

- The instruments are:

$$Z^H_{at} = \sum_o s_{ao} \times (c_{ot} - \bar{c}_t) \times H_a, \qquad Z^L_{at} = \sum_o s_{ao} \times (c_{ot} - \bar{c}_t) \times L_a$$

Heterogeneity by market integration: first stage

	IDP flow within 25km × High market integration	**IDP flow within 25km × Low market integration**
Recentered shock	0.25***	0.26***
	(0.08)	(0.05)
BH 2023 RI p-value	0.081	0.010
Cluster-robust Wald test of coefficient equality		
Coefficient (High − Low)	-0.01	
p-value	0.928	
Wald test degrees of freedom	1, 96	
Sanderson-Windmeijer F	7.26	8.68
SW F p-value	0.008	0.004
SW F degrees of freedom	1, 96	1, 96
IDP flow sample mean (persons)	2,933	1,920
Origin-year cells (shifts)	67	158
Effective shifts	3.08	24.01
Origin-woreda clusters	38	87

Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

If low-skill labor supply effects dominate demand effects:

High market integration	Low market integration
1. Low-skill to high-skill wage ratio falls	1. –
2. Hosts move towards high-skill occupation	2. Limited reallocation
3. Prices for local services fall	3. –
4. Food prices are unchanged	4. Prices adjust locally

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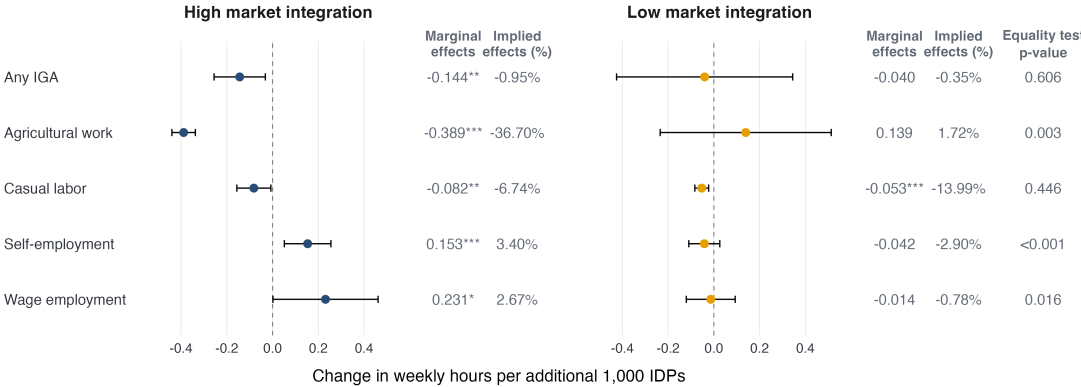
High market integration

1. Low-skill to high-skill wage ratio falls
2. Hosts move towards high-skill occupation
3. Prices for local services fall
4. Food prices are unchanged

Low market integration

1. –
2. **Limited reallocation**

Limited reallocation with low market integration



Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

If low-skill labor supply effects dominate demand effects:

High market integration

1. Low-skill to high-skill wage ratio falls
2. Hosts move towards high-skill occupation
3. Prices for local services fall
4. Food prices are unchanged

Low market integration

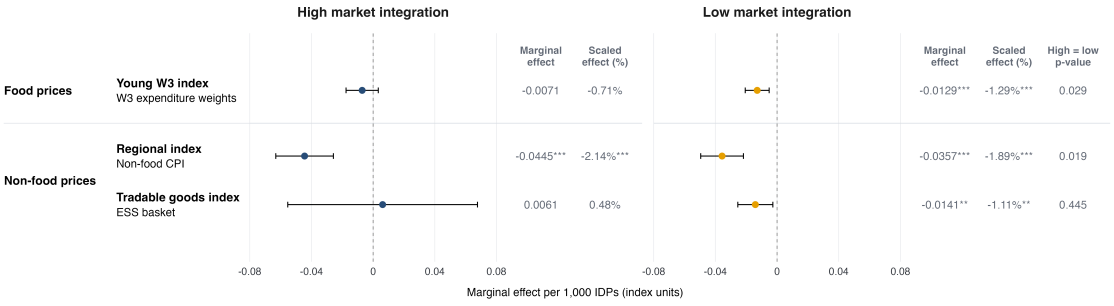
1. –
2. **Limited reallocation**
   No evidence of reallocation

Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

If low-skill labor supply effects dominate demand effects:

High market integration	Low market integration
1. Low-skill to high-skill wage ratio falls	1. –
2. Hosts move towards high-skill occupation	2. **Limited reallocation** No evidence of reallocation
3. Prices for local services fall	3. –
4. Food prices are unchanged	4. **Prices adjust locally**

With low integration, prices adjust locally



Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

If low-skill labor supply effects dominate demand effects:

High market integration	Low market integration
1. Low-skill to high-skill wage ratio falls	1. –
2. Hosts move towards high-skill occupation	2. **Limited reallocation** No evidence of reallocation
3. Prices for local services fall	3. –
4. Food prices are unchanged	4. **Prices adjust locally** Stronger price reductions in low-integration areas

Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

Two first-order empirical questions remain:

1. Do host communities benefit from displacement in terms of consumption?

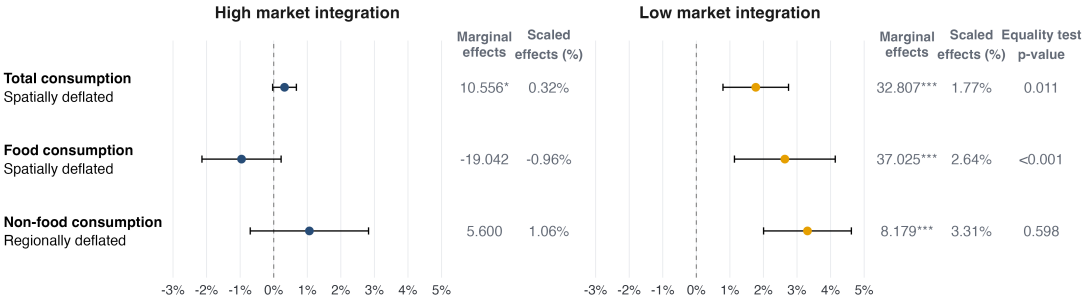
2. How does agriculture adjust?

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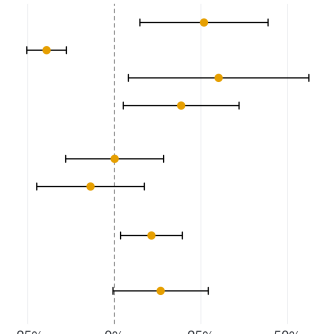
Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

Two first-order empirical questions remain:

1. **Do host communities benefit from displacement in terms of consumption?**
   Largest increase in low-integration areas

2. **How does agriculture adjust?**

How does agriculture adjust?



		Marginal effects	Outcome mean	Scaled effects
Farm labor	Family labor person-days	5.706***	22.049	25.88%
	Hired labor person-days	-1.178***	6.017	-19.57%
	Unpaid labor person-days	2.181**	7.245	30.11%
	Total labor person-days	**6.813****	**35.322**	**19.29%**
Host unpaid labor	Any unpaid labor	0.000	0.154	0.08%
	Households worked for	-0.038	0.553	-6.88%
Land	Cultivated hectares	0.051**	0.473	10.71%
Production	Production value at contemporaneous prices	210.949*	1578.387	13.36%

Units: farm labor in person-days/household/season; host unpaid labor per adult, past 12 months; land in ha; production in 2021 PPP USD.
Scaled effects: % of outcome mean per additional 1,000 IDPs. Orange: low integration. Whiskers: 95% BHJ intervals.

Does the effect of displacement on the labor and consumption markets of host communities differ by market integration?

Two first-order empirical questions remain:

1. **Do host communities benefit from displacement in terms of consumption?**
   Largest increase in low-integration areas

2. **How does agriculture adjust?**
   Farm labor and cultivated land increase in low-integration areas

Conclusion

1) Displacement is a labor supply and a consumer demand shock

- In Ethiopia, labor-supply shock dominates consumer-demand shock

2) The effect of displacement on host communities differs by market integration

- **High integration**: hosts reallocate toward nonfarm employment
- **Low integration**: expansion within agriculture & lower prices

Implications for policy and development

1. Displaced people in Ethiopia were valued workers and had positive impact on host communities
2. Hosts' reallocation out of agriculture only takes place where markets are integrated
   $\Rightarrow$ Structural transformation?
3. Gains in living standards can accompany expansion within agriculture

Thank you!

Feedback welcome

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